

VIHTI MAASOJA, Finland

WMO: 027580

Lat: 60.419N

Lon: 24.399E

Elev: 145

StdP: 14.62

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.8	-6.8	-19.1	1.9	-14.8	-10.5	3.1	-6.5	17.6	26.9	15.2	27.6	0.1	180	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	81.9	66.9	78.2	64.7	75.0	62.6	69.6	78.3	67.1	75.3	64.8	72.1	7.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	98.0	73.5	64.1	90.0	70.7	61.9	83.3	68.1	33.7	78.4	31.7	75.3	29.9	72.2	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	12.8	11.2	-16.7	84.9	11.6	4.3	-25.0	88.0	-31.8	90.5	-38.3	92.9	-46.6	96.1
			-16.0	71.6	11.5	2.6	-24.3	73.5	-31.0	75.0	-37.5	76.5	-45.9	78.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.9	19.8	22.9	27.9	38.6	49.8	56.5	61.7	58.7	51.2	40.5	34.7	27.0	(6)
	DBStd	16.25	12.54	12.02	9.25	6.04	7.38	5.91	5.67	5.18	5.82	6.50	7.96	10.41	(7)
	HDD50	4361	937	758	685	343	95	13	0	3	55	301	459	712	(8)
	HDD65	8860	1402	1178	1150	792	473	263	136	205	415	760	909	1177	(9)
	CDD50	1029	0	0	0	2	89	206	364	273	90	5	0	0	(10)
	CDD65	51	0	0	0	0	1	6	34	10	0	0	0	0	(11)
	CDH74	914	0	0	0	0	75	167	510	161	1	0	0	0	(12)
	CDH80	169	0	0	0	0	8	20	111	30	0	0	0	0	(13)
	WSAvg	4.5	4.0	4.4	5.0	4.6	4.9	5.1	4.3	4.0	4.0	4.5	4.5	4.6	(14)
	PrecAvg	27.3	2.5	1.7	1.5	1.3	1.6	2.6	2.9	3.1	2.1	3.3	2.9	2.3	(15)
(16) Precipitation	PrecMax	33.2	4.3	4.2	2.6	2.3	3.5	4.4	8.0	6.6	5.1	7.7	6.8	4.1	(16)
	PrecMin	21.0	0.4	0.1	0.2	0.2	0.3	0.9	0.0	0.4	0.4	1.2	0.4	0.3	(17)
	PrecStd	3.4	0.9	1.1	0.6	0.6	0.9	1.1	1.9	1.5	1.3	1.5	1.5	1.0	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.1	42.2	51.1	67.4	80.7	82.8	86.7	84.1	71.8	57.8	51.0	47.5	(19)
		MCWB	38.6	37.8	42.4	51.4	61.3	64.9	69.2	67.3	62.5	53.9	49.5	46.7	(20)
	2%	DB	38.3	39.1	45.6	62.2	75.5	78.7	83.1	78.0	67.6	54.9	48.6	43.7	(21)
		MCWB	37.4	37.3	39.6	48.6	58.9	62.9	68.7	65.5	60.5	52.3	47.9	42.4	(22)
	5%	DB	36.4	37.0	41.8	57.0	71.6	75.1	79.7	74.5	64.8	53.2	46.3	41.0	(23)
		MCWB	35.8	36.0	37.5	46.4	56.8	61.6	66.9	63.1	58.2	51.1	45.4	40.0	(24)
	10%	DB	34.3	35.5	38.8	52.2	67.1	70.9	76.2	71.1	62.2	51.3	44.6	38.5	(25)
		MCWB	33.8	34.6	35.9	43.3	54.6	59.3	65.2	61.5	57.7	49.4	43.7	37.6	(26)
	0.4%	WB	39.0	39.1	43.2	52.8	64.3	69.0	72.2	71.4	63.9	55.7	49.9	46.9	(27)
		MCDB	39.6	40.6	47.8	64.5	75.0	77.9	80.7	78.8	68.9	57.0	50.6	47.4	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.6	37.6	40.3	50.0	61.2	65.6	70.3	67.5	61.8	53.6	48.0	42.9	(29)
		MCDB	38.2	38.8	44.4	60.8	70.4	75.1	79.2	74.3	66.0	54.5	48.8	43.7	(30)
	5%	WB	36.0	36.1	38.3	47.1	58.8	63.1	68.8	65.2	60.0	51.8	45.7	40.2	(31)
		MCDB	36.5	37.1	41.1	54.9	69.6	72.1	77.6	71.6	63.4	52.8	46.2	41.1	(32)
	10%	WB	33.9	34.8	36.4	44.5	55.8	60.8	66.5	63.4	58.3	49.6	43.9	37.7	(33)
		MCDB	34.3	35.5	38.4	50.9	65.2	68.9	74.6	69.0	61.2	50.9	44.6	38.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	12.2	16.3	19.6	23.5	20.8	21.7	21.4	18.0	13.3	8.1	8.9	(35)
		MCDBR	7.0	9.5	17.7	30.8	32.5	29.5	27.6	28.1	22.5	15.9	8.9	9.0	(36)
		MCWBR	6.7	8.1	11.8	17.9	16.8	15.3	14.6	16.3	15.2	13.0	8.5	8.5	(37)
	5% WB	MCDBR	6.8	8.7	14.9	26.6	27.7	24.2	24.8	22.4	19.3	12.2	8.9	8.5	(38)
		MCWBR	6.6	7.6	10.8	16.8	15.0	13.6	13.9	13.9	14.0	11.0	8.7	8.3	(39)
(40) Clear-Sky Solar Irradiance	taub		0.272	0.309	0.332	0.361	0.358	0.355	0.378	0.370	0.342	0.331	0.300	N/A	(40)
	taud		2.125	2.189	2.258	2.357	2.420	2.459	2.415	2.435	2.506	2.398	2.213	N/A	(41)
	Ebn at Noon		149	213	250	265	275	278	267	260	249	207	138	N/A	(42)
	Edh at Noon		13	23	29	32	32	32	32	30	23	19	12	N/A	(43)
(44) All-Sky Solar Radiation	RadAvg		78	259	730	1218	1648	1713	1653	1277	779	341	96	46	(44)
	RadStd		14	54	94	125	176	137	154	134	102	56	18	7	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (59 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	-248	N/A	N/A	N/A	

Nomenclature: See separate page